Module: external integrations (e-mail, Telegram, snapshot)

English version of ../../modules/integrations.md.

1. Purpose and boundaries

Optional channels: SMTP e-mail, the Telegram Mini App/bot, and snapshot ingestion from a source. Every channel is enabled separately and does not affect the analytics. It is not a mandatory data path and never blocks the start.

2. Use scenarios

- Sending a report e-mail (`integrations/email.py`).
- Signing in to the Mini App through the Telegram signature and a one-time code.
- Ingesting a new snapshot by token.
- Failure: a channel is not configured → the operation reports "not configured" and the service keeps working.

3. Contract

`/api/v1/email/*`, `/api/v1/telegram/*`, `PUT /api/v1/source/snapshot` — in `backend/openapi.json`. Telegram providers — `telegram/providers.py`.

4. Data

No tables of its own. Telegram sessions and OTP live in memory or a cache with TTL (`AEZA_TELEGRAM_SESSION_TTL_SECONDS`, `AEZA_TELEGRAM_OTP_TTL_SECONDS`); this is deliberate: an OTP must not survive a restart.

5. Access rules

The snapshot — `AEZA_SOURCE_INGEST_TOKEN` (constant-time comparison). Telegram — signature verification and maximum data age (`AEZA_TELEGRAM_AUTH_MAX_AGE_SECONDS`), plus an OTP attempt limit. E-mail — authenticated users only.

6. Dependencies

Outgoing: an SMTP server, the Telegram Bot API. External systems are isolated behind adapters, so replacing a provider does not touch the domain.

7. Configuration

`AEZA_TELEGRAM_BOT_ENABLED`, `AEZA_TELEGRAM_BOT_TOKEN`, `AEZA_TELEGRAM_WEBHOOK_SECRET`, `AEZA_TELEGRAM_WEBHOOK_URL`, `AEZA_TELEGRAM_MINI_APP_URL`, `AEZA_TELEGRAM_OTP_*`, `AEZA_SOURCE_INGEST_TOKEN`, SMTP settings. An empty token means the channel is off.

8. Build and debug

```
python -m pytest backend/tests/test_email.py -q
python -m pytest backend/tests -q -k telegram
```

9. Tests

`test_email.py` (including `aiosmtpd` scenarios), the Telegram tests in `backend/tests`, and `test_api.py` for the snapshot routes. Refusal without configuration is checked explicitly.

10. Observability

Channel errors are logged without secrets (enforced by `check_docs.py` for "secret-like" examples and by the logging code). Operator action: check that tokens exist and that the external service is reachable.

11. Limitations and readiness

- Telegram session state is in memory: several replicas would need shared storage (not supported today).
- No retry queue for e-mail.
- Readiness: any channel can be switched off and the product keeps working.